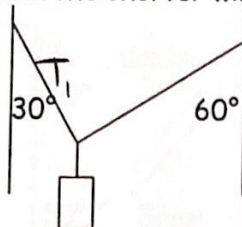


Equilibrium Worksheet#2

(name) _____

1. A lamp is supported between two vertical poles with two wires attached to the poles as shown. The lamp hangs from a short chain. If the lamp has a mass of 11 kg, what is the tension in the shorter wire?



$T_1 =$ _____.

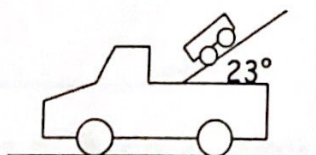
2. As an airplane accelerates along a horizontal runway, a pendulum hanging from its ceiling is displaced 12° from the vertical. Calculate the acceleration of the plane.

$a =$ _____.

3. A uniform ladder has a mass of 16 kg and a length of 8.5 m. It stands on the ground and leans against a vertical wall, making an angle of 62° with the ground. The friction between the ladder and the wall is not significant. What is the smallest force of friction between the ladder and the ground which will just prevent the ladder from slipping?

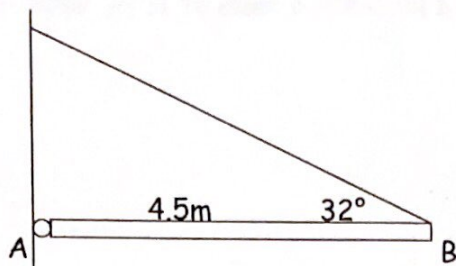
$F_f =$ _____.

4. The following diagram shows a cart on a ramp which is fixed to the floor in the back of a truck. Friction in the wheels of the cart is negligible. At what rate must the truck accelerate forward for the cart to remain stationary with respect to the ramp?



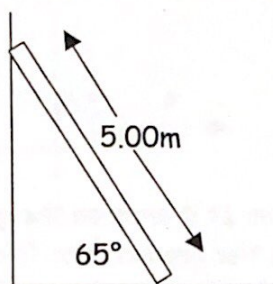
$a =$ _____.

5. The following diagram shows a uniform 24 kg beam hinged to the wall at A and supported by a cable from B. What is the magnitude of the horizontal force exerted by the wall on the beam?



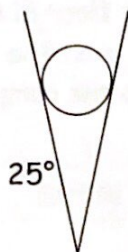
$F_x =$ _____.

6. A uniform ladder with a mass of 15 kg rests against a wall as shown. If the friction from the wall is not significant, what is the minimum coefficient of friction between the ladder and the floor that will keep the ladder from slipping?



$\mu =$ _____.

7. A steel ball of mass 1.7 kg rests in the V-shaped wedge formed by two steel plates as shown in the diagram. Find the magnitude of the force exerted on each plate by the ball if the friction between the ball and the plates is not significant.



$F =$ _____.

Answers: 1. 93N 2. 2.1m/s^2 3. ~~41.7N~~
42N 4. ~~4.16m/s²~~
4.2m/s² 5. ~~188N~~
190N 6. 0.23 7. 38N